

Technical Document #57: Cybersecurity - Comprehensive Overview

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Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Incident response plans define how organizations respond to security incidents. They include preparation, detection, containment, and recovery phases.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Key Takeaways:

- Social engineering manipulates people into revealing confidential information.
- Security information and event management (SIEM) systems collect and analyze security logs.
- Intrusion detection systems (IDS) monitor network traffic for suspicious activity.